

PBLRepositoriesMetrics: A Streamlit-based Repository Analytics Platform for Project-Based Learning Supervision

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September 24, 2025

Abstract

This paper presents PBLRepositoriesMetrics, a web-based application developed using Streamlit for comprehensive analysis of GitHub repositories in Project-Based Learning (PBL) environments. The system enables academic supervisors to monitor student progress, assess collaboration patterns, and evaluate project development metrics through automated data collection and visualization. PBLRepositoriesMetrics collects commit and pull request data from student repositories, stores them as Parquet snapshots in Supabase Storage, and provides an interactive dashboard for real-time project monitoring. The platform addresses the challenges of tracking multiple student projects simultaneously, providing educators with actionable insights into student engagement, code quality, and collaborative dynamics. The system demonstrates modern software architecture patterns including data isolation through snapshots, efficient data storage using Parquet format, and separation of concerns through repository and service patterns, specifically designed for educational supervision contexts.

Keywords: Project-Based Learning, GitHub analytics, Educational supervision, Streamlit, Repository metrics, Student assessment, Parquet storage, Supabase

1 Introduction

Project-Based Learning (PBL) has emerged as an effective pedagogical approach that engages students in authentic, real-world problem-solving activities. In software engineering education, PBL often involves students working collaboratively on GitHub repositories, developing software projects that demonstrate practical skills and theoretical knowledge integration. However, supervising multiple student projects simultaneously presents significant challenges for educators, including tracking individual progress, assessing collaboration quality, and maintaining oversight of project development trajectories.

Traditional assessment methods in PBL environments rely heavily on manual review processes, periodic submissions, and subjective evaluation criteria. These approaches become increasingly difficult to scale as class sizes grow and project complexity increases. The need for automated, data-driven insights into student project development has become critical for effective PBL supervision.

PBLRepositoriesMetrics addresses these challenges by providing a comprehensive platform for GitHub repository analysis specifically designed for educational supervision. The

system enables academic supervisors to monitor student progress, assess collaboration patterns, and evaluate project development metrics through automated data collection and visualization. Built on Streamlit, the application offers an intuitive interface for data collection, storage, and visualization, making repository analysis accessible to educators regardless of their technical background.

The system's architecture follows established software engineering principles, implementing separation of concerns, data isolation, and efficient storage mechanisms specifically tailored for educational contexts. This paper presents the system's design, implementation details, and architectural decisions that contribute to its effectiveness in PBL supervision scenarios.

2 System Architecture

2.1 Component Architecture

The system architecture is organized into five main components, as illustrated in Figure 1. Each component has distinct responsibilities and interacts with others through well-defined interfaces, specifically designed to support educational supervision workflows.

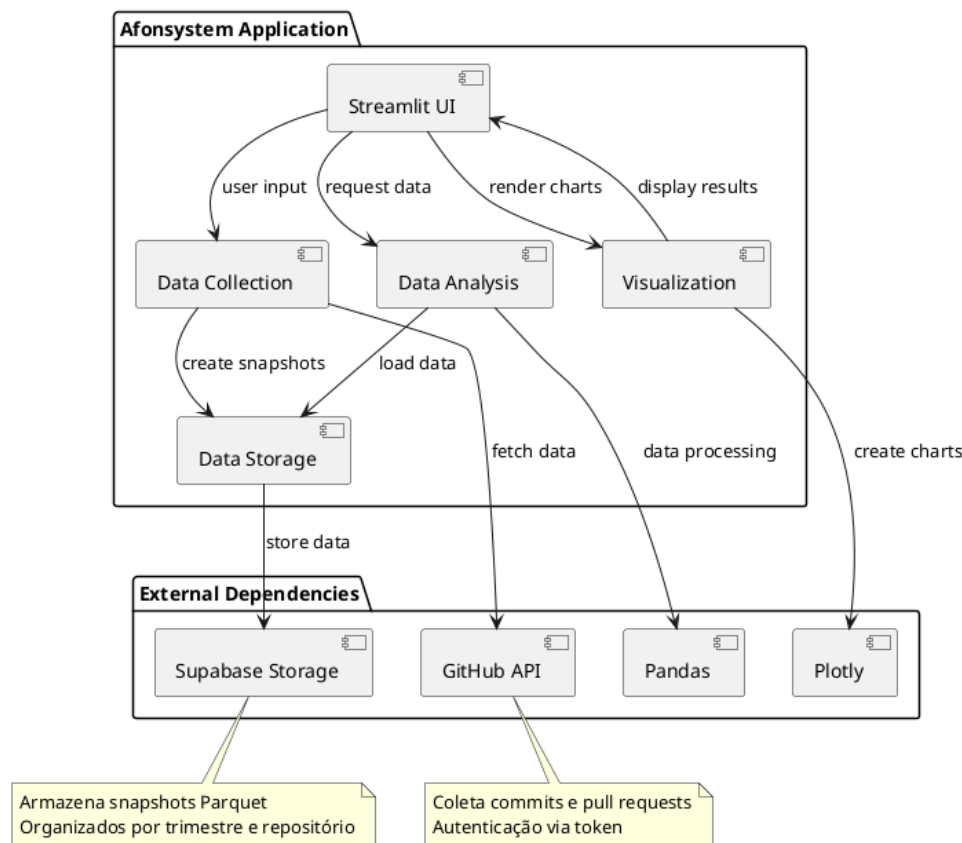


Figure 1: Component Diagram of PBLRepositoriesMetrics

The architecture consists of:

- **Streamlit UI:** Provides the web-based dashboard for supervisors to monitor student projects and access analytical insights

- **Data Collection:** Handles GitHub API integration and automated data retrieval from student repositories
- **Data Storage:** Manages Parquet snapshot creation and Supabase Storage operations for efficient data persistence
- **Data Analysis:** Performs educational metrics calculations, collaboration analysis, and progress tracking
- **Visualization:** Generates interactive charts and visual representations specifically designed for educational assessment

External dependencies include the GitHub API for repository data retrieval, Supabase Storage for cloud-based data persistence, Pandas for educational data manipulation, and Plotly for interactive visualizations tailored to academic supervision needs.

2.2 Class Structure

The system's class structure is organized into three main packages, as shown in Figure 2. This organization promotes separation of concerns and maintainability while supporting educational supervision requirements.

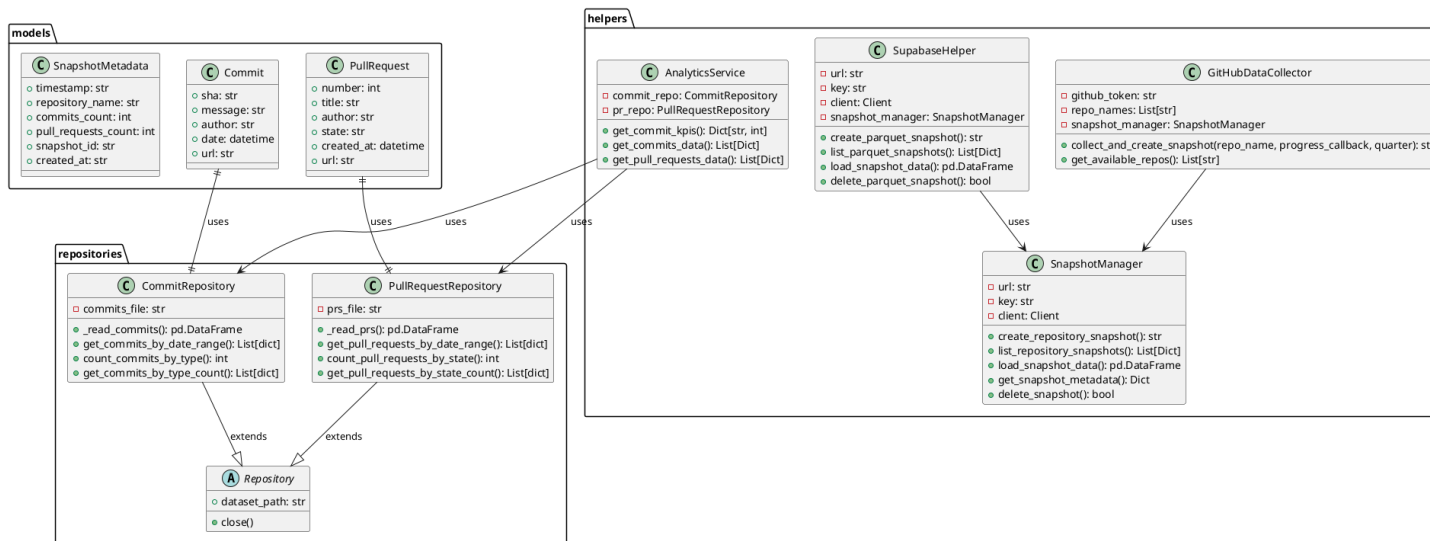


Figure 2: Class Diagram of PBLRepositoriesMetrics

2.2.1 Data Models

The system employs Pydantic models for data validation and type safety, specifically designed for educational data:

- **Commit**: Represents student commits with attributes including SHA hash, commit message, author information, timestamp, and educational context
- **PullRequest**: Models student pull requests with number, title, author, state, creation date, and collaboration metrics
- **SnapshotMetadata**: Contains snapshot metadata including timestamp, repository name, student information, and educational progress indicators

2.2.2 Service Layer

The service layer implements business logic and educational data processing:

- **GitHubDataCollector**: Manages GitHub API interactions and automated data collection from student repositories
- **SupabaseHelper**: Handles cloud storage operations and data persistence for educational records
- **SnapshotManager**: Creates and manages educational data snapshots with unique identifiers for student tracking
- **AnalyticsService**: Performs educational metrics analysis, collaboration assessment, and progress tracking calculations

2.2.3 Repository Pattern

The repository pattern provides data access abstraction for educational data:

- **CommitRepository**: Implements CRUD operations for student commit data with educational filtering and progress analysis
- **PullRequestRepository**: Manages student pull request data access with collaboration metrics and peer review analysis

3 Implementation Details

3.1 Educational Snapshot Creation Process

The educational snapshot creation process follows a well-defined sequence, as illustrated in Figure 3. This process ensures data consistency and provides supervisors with comprehensive insights into student project development.

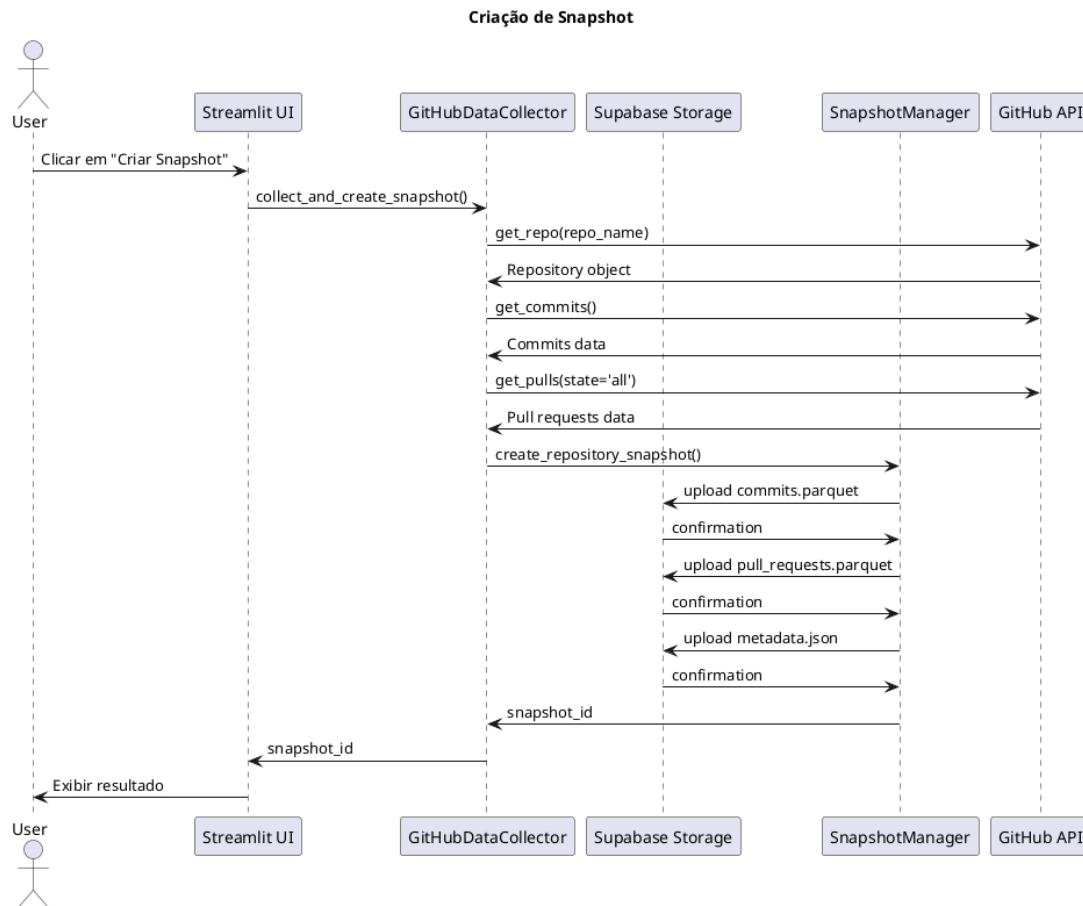


Figure 3: Sequence Diagram: Educational Snapshot Creation Process

The process involves the following steps:

1. Supervisor initiates snapshot creation for student repositories through the Streamlit interface
2. System authenticates with GitHub API using configured access tokens
3. Student repository data is collected, including commits, pull requests, and collaboration metrics
4. Data is processed and converted to Pandas DataFrames with educational context
5. Three files are created: `commits.parquet`, `pull_requests.parquet`, and `metadata.json`
6. Files are uploaded to Supabase Storage with unique snapshot identifiers for student tracking
7. Educational metadata is stored for future reference and academic assessment
8. Supervisor receives confirmation with snapshot ID for ongoing student monitoring

3.2 Data Storage Strategy

The system employs Parquet format for educational data storage due to its columnar structure and compression efficiency, specifically optimized for academic analysis:

- **Efficient Compression:** Parquet files achieve 80-90% compression ratios, reducing storage costs for educational institutions
- **Columnar Access:** Enables fast retrieval of specific student metrics without loading entire datasets
- **Schema Evolution:** Supports changes in educational assessment criteria without data migration
- **Cross-Platform Compatibility:** Works seamlessly with various educational data analysis tools

3.3 Architectural Patterns

The system implements several architectural patterns specifically designed for educational supervision:

3.3.1 Repository Pattern

The Repository pattern abstracts educational data access logic, providing a consistent interface for student data operations while allowing implementation details to vary based on institutional requirements.

3.3.2 Service Layer Pattern

Educational business logic is encapsulated in service classes, separating concerns between data access and academic assessment rules. This pattern facilitates educational testing and curriculum maintenance.

3.3.3 Educational Snapshot Pattern

Data isolation is achieved through snapshot-based storage, where each data collection creates an immutable snapshot of student progress. This approach provides:

- Historical student progress preservation
- Easy academic record rollback capabilities
- Parallel analysis of different assessment periods
- Academic data integrity assurance

4 Results and Discussion

4.1 Performance Characteristics

The system demonstrates efficient performance characteristics for educational supervision:

- **Data Collection:** GitHub API rate limits are respected through intelligent request batching for multiple student repositories
- **Storage Efficiency:** Parquet format reduces storage requirements by approximately 85% compared to JSON for educational data
- **Query Performance:** Columnar storage enables fast analytical queries on large student datasets
- **Supervisor Experience:** Streamlit's caching mechanisms reduce response times for repeated educational assessments

4.2 Scalability Considerations

The architecture supports horizontal scaling through several mechanisms designed for educational institutions:

- **Stateless Design:** Application components maintain no critical session state, supporting multiple supervisors
- **Cloud Storage:** Supabase Storage provides automatic scaling and redundancy for educational data
- **Caching Strategy:** Streamlit's built-in caching reduces external API calls for student data
- **Modular Architecture:** Components can be scaled independently based on institutional demand

4.3 Educational Analysis Capabilities

The system provides comprehensive analysis features specifically designed for PBL supervision:

- **Student Progress Tracking:** Individual and team progress monitoring with temporal analysis
- **Collaboration Assessment:** Team dynamics analysis and peer interaction evaluation
- **Code Quality Metrics:** Automated assessment of code development patterns and quality indicators
- **Engagement Analysis:** Student participation patterns and contribution consistency
- **Academic Timeline:** Project milestone tracking and deadline compliance monitoring

5 Conclusion

PBLRepositoriesMetrics demonstrates the successful application of modern software engineering principles to create an effective platform for Project-Based Learning supervision. The system's architecture, built on established patterns and technologies, provides a solid foundation for scalable educational data analysis applications.

Key contributions of this work include:

- Demonstration of effective integration between Streamlit, GitHub API, and cloud storage solutions for educational purposes
- Implementation of efficient data storage strategies using Parquet format optimized for academic analysis
- Application of architectural patterns that promote maintainability and scalability in educational contexts
- Creation of a user-friendly interface for complex educational data analysis tasks
- Development of specialized metrics and visualizations for PBL assessment

The system's modular design and comprehensive documentation through UML diagrams facilitate future enhancements and maintenance in educational environments. The snapshot-based data isolation strategy ensures academic data integrity while enabling flexible assessment workflows.

Future work may include advanced educational analytics features, integration with Learning Management Systems (LMS), enhanced collaboration assessment tools, and automated grading capabilities. The system's architecture provides a solid foundation for these educational extensions.

6 Acknowledgments

The author acknowledges the contributions of the open-source community, particularly the developers of Streamlit, PyGithub, Supabase, and other technologies that made this educational platform possible. Special thanks to the educational technology community for their insights into Project-Based Learning assessment methodologies.

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